

AnalogForge Replacements And Variables

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Purpose

This guide lets an operator run the Analog Forge kit even when the exact suggested tool is unavailable. The replacement rule is:

preserve function first
preserve color second
preserve physical form third

Do not stop a run because a perfect material is missing. Replace the tool with something that performs the same state role and receipt the substitution.

Replacement Receipt

Every substitution should be recorded:

original suggested tool
replacement tool
preserved function
lost function, if any
color mapping
boundary effect
operator initials
receipt id

Universal Replacement Table

Suggested tool | Primary function | Acceptable replacements

Grey loose paper | unresolved substrate | printer paper, index card, cardboard, whiteboard zone, tablet blank page
Colored paper | state domain | colored sticky note, colored tape boundary, marker fill, labeled card
Gradient page | relation between states | two or three colored stripes, layered sticky notes, clear overlay with color marks
Notebook | active state archive | binder section, folder, envelope, digital folder, labeled box
Binder | large closure archive | file box, project folder, zip pouch, cabinet drawer, digital repository
C token | current center | coin, button, poker chip, sticker, magnet, drawn circle, small stone
Tokens | movable local bodies | coins, beads, washers, LEGO pieces, buttons, paper squares
Strings | edge, bond, braid | yarn, thread, tape line, drawn line, pipe cleaner, wire, rubber band
Stickers | fixed state marker | tape flag, marker dot, stamp, clipped paper tab, label
Notecard | decision probe | sticky note, scrap card, digital card, index strip, folded paper
Clear sleeve | reversible overlay | transparency sheet, tracing paper, glass pane, laminated sheet, tablet layer
Dice | bounded uncertainty | coin flips, spinner, numbered slips, random card draw, random number table
Playing cards | permutation/order | numbered cards, slips in a cup, spreadsheet order, dice sequence
Balsa | rigid lattice edge | skewers, straws, coffee stirrers, chopsticks, wire, craft sticks
Tabs/dividers | navigation | sticky tabs, binder clips, colored tape, folder labels
Receipt sheet | recovery record | notebook page, form printout, markdown file, JSON receipt

Color Replacement

If exact colors are missing, preserve color role with labels.

Role | Preferred | Replacement

red | red material | label RED, triangle mark, R symbol
green | green material | label GREEN, square mark, G symbol
blue | blue material | label BLUE, circle mark, B symbol
white | white material | label PROOF, open circle, W symbol
black | black material | label OBLIGATION, filled square, K symbol
grey | grey material | label UNRESOLVED, dotted border
clear | clear material | label OVERLAY, transparent or dashed border
neon | neon material | label BOUNDARY, starburst mark

If You Lack Eight Copies

Eight copies per color is the ideal. If you do not have all eight, use slot labels:

slot 1
slot 2
slot 3
slot 4
slot 5
slot 6
slot 7
slot 8

The important thing is not eight physical luxury objects. The important thing is that the operator does not collapse eight possible local readout positions into one unmarked object.

If You Lack A Physical Tool Entirely

Use a drawn version.

drawn token = circle with ID
drawn string = line with source/target
drawn sticker = boxed label
drawn dice = recorded branch sampler
drawn card = rank/suit written on paper
drawn balsa = edge segment with endpoints

Drawn tools are valid if the receipt states they are drawn replacements.

Low-Cost Kit

Minimum low-cost setup:

printer paper
colored pens
index cards
string or yarn
coins or buttons
standard card deck
one die
clear plastic sleeve
folder or binder
receipt template

Classroom Kit

Use durable replacements:

- laminated boards
- dry erase markers
- magnetic tokens
- magnetic strings or tape
- reusable card decks
- scannable receipt sheets
- student binders

Field Kit

For outdoor or quick experimental work:

- clipboard
- waterproof notebook
- colored tape
- survey flags
- stones or washers
- cord
- laminated receipt cards
- phone camera
- zip pouch

Lab Kit

For scientific work:

- cataloged tokens
- calibrated colors
- registered grid boards
- QR-coded tools
- controlled dice/cards
- photo station
- digital receipt mirror
- chain-of-custody log

Replacement Safety Rule

If a replacement changes the boundary, mark it in neon and receipt it.

Examples:

- rubber band instead of string -> elasticity added
- marker line instead of string -> relation no longer movable
- paper token instead of magnet -> token may drift
- phone photo instead of receipt scan -> image quality obligation

The replacement is still valid, but its changed behavior becomes part of the run.

Completion Rule

A replacement is valid when:

- its function is named
- its color role is named
- its lost behavior is named
- its receipt exists

Without those four records, the replacement remains informal.